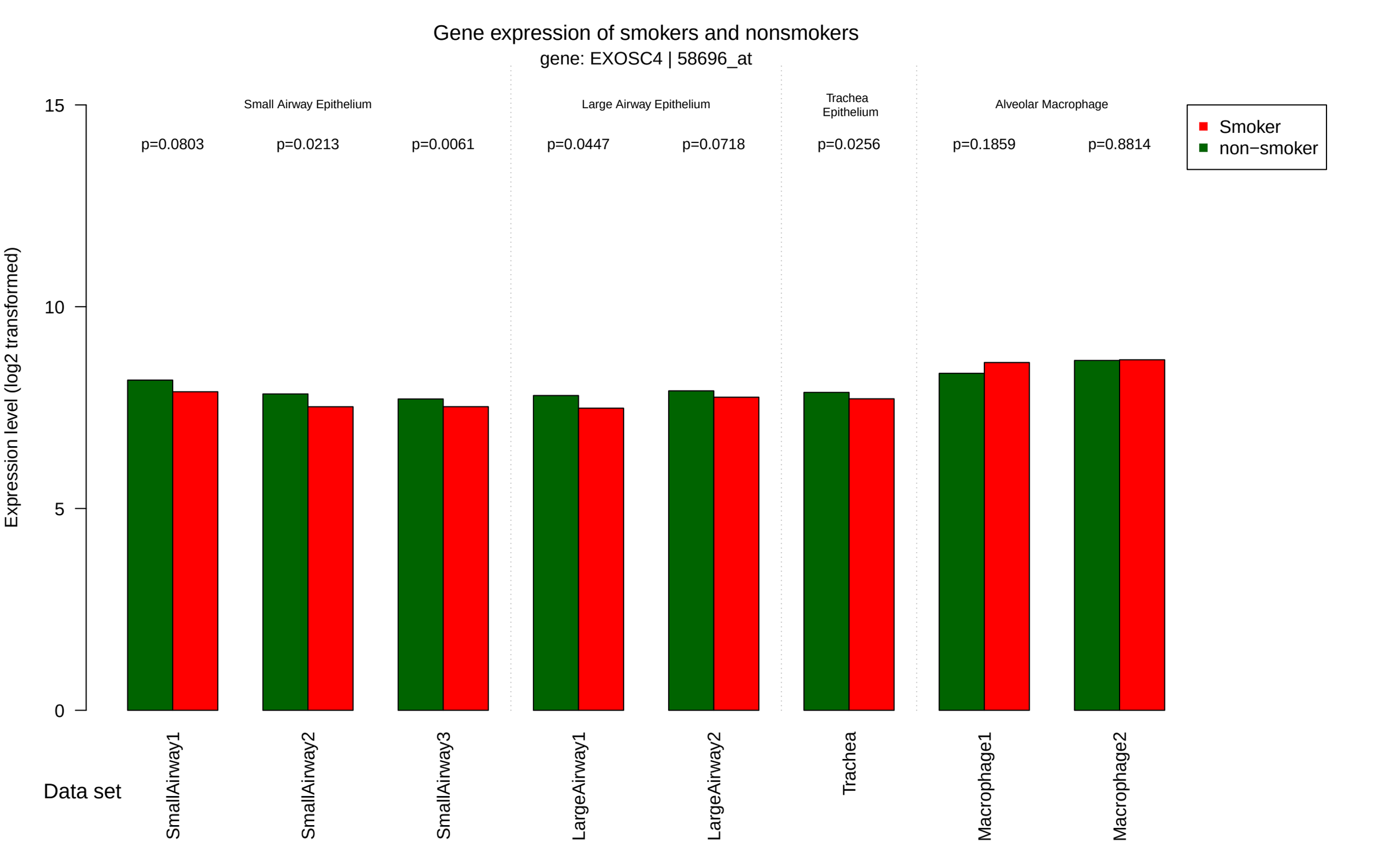
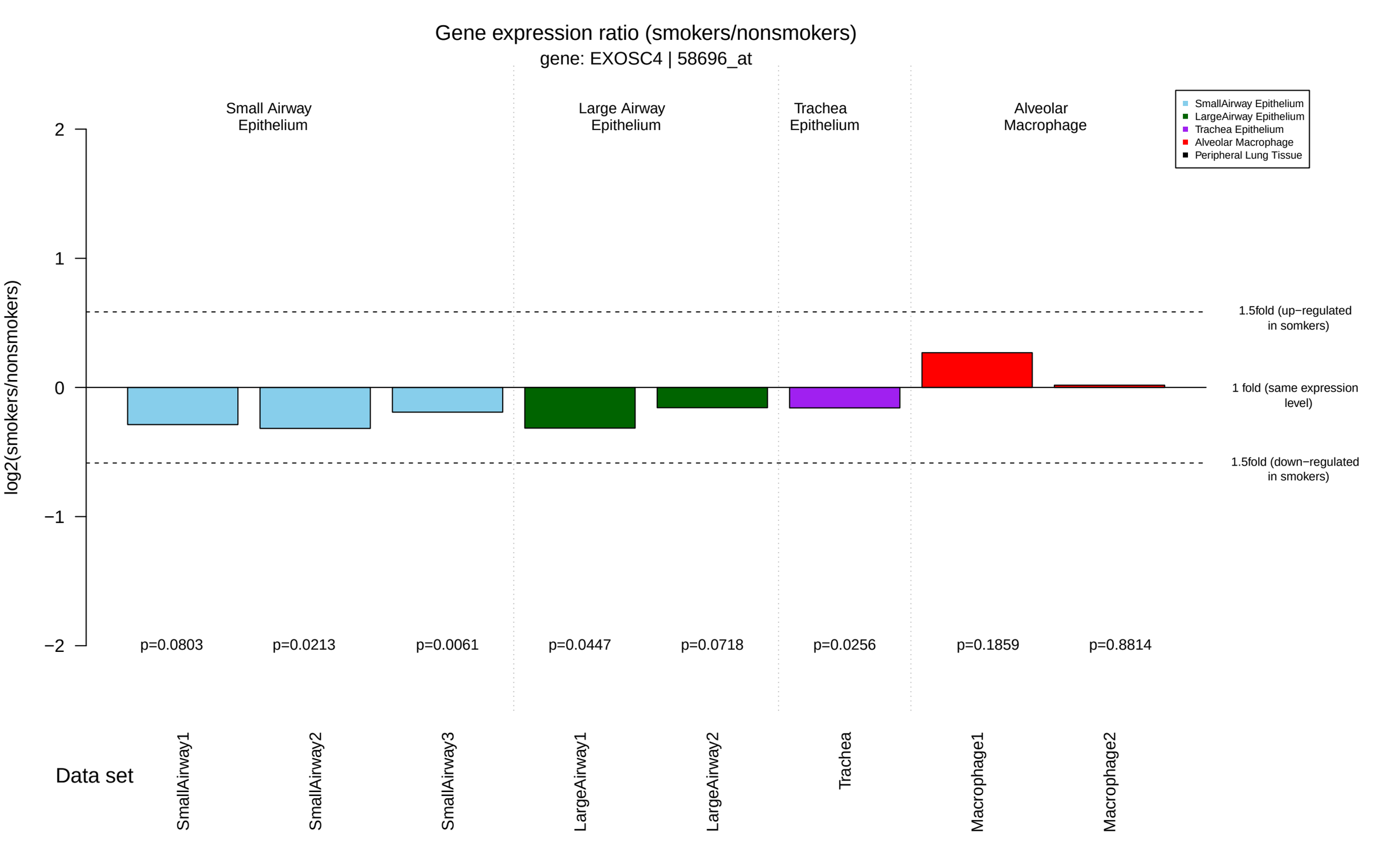
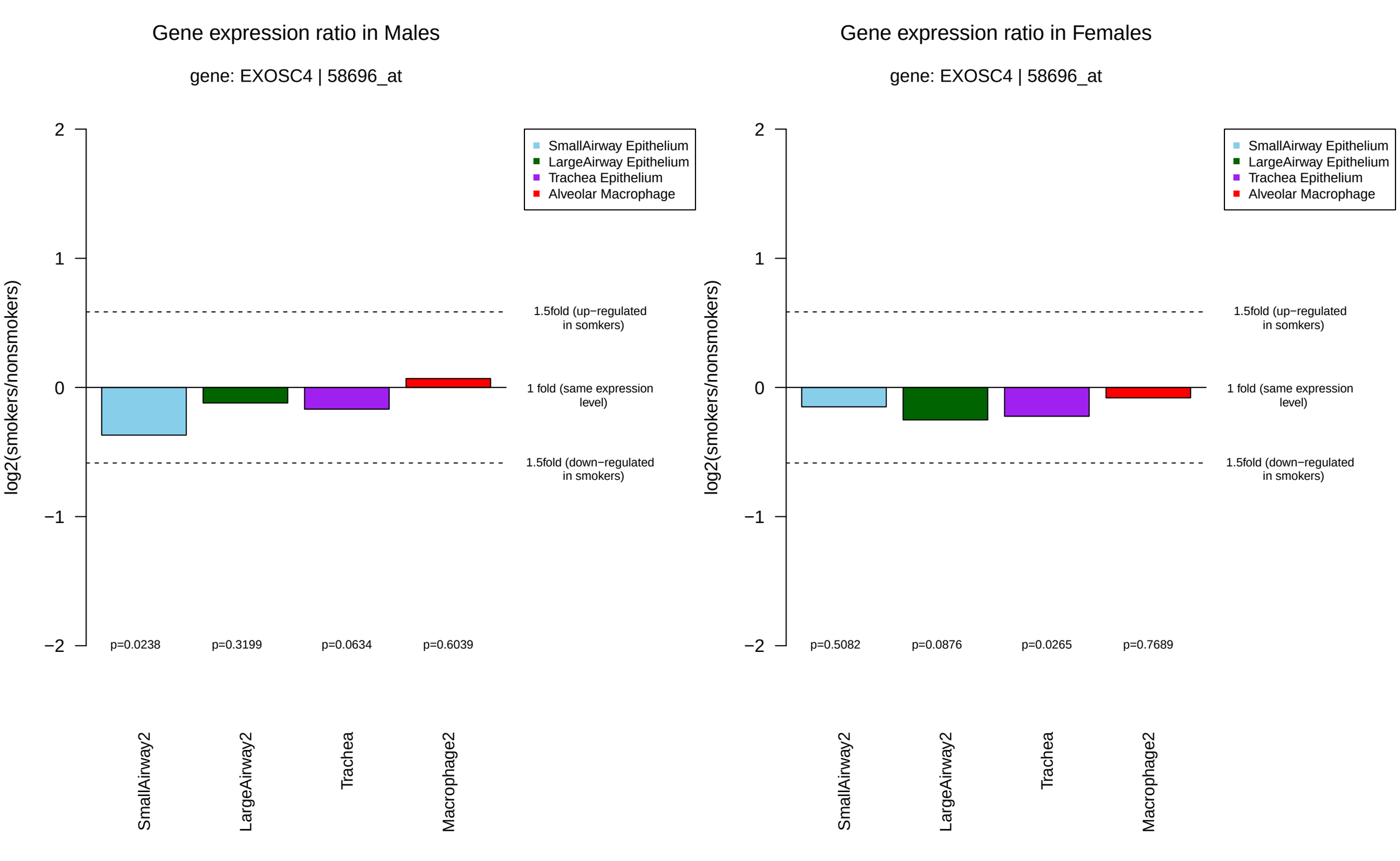
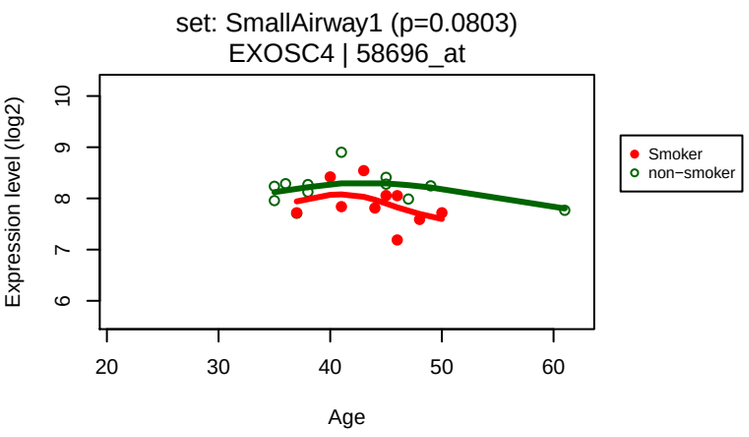
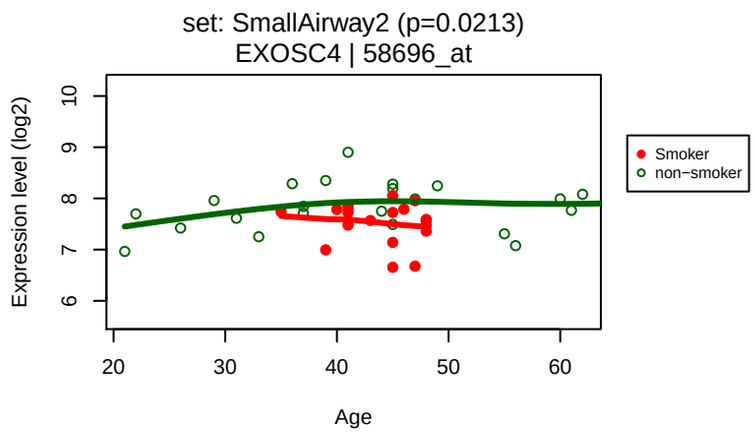
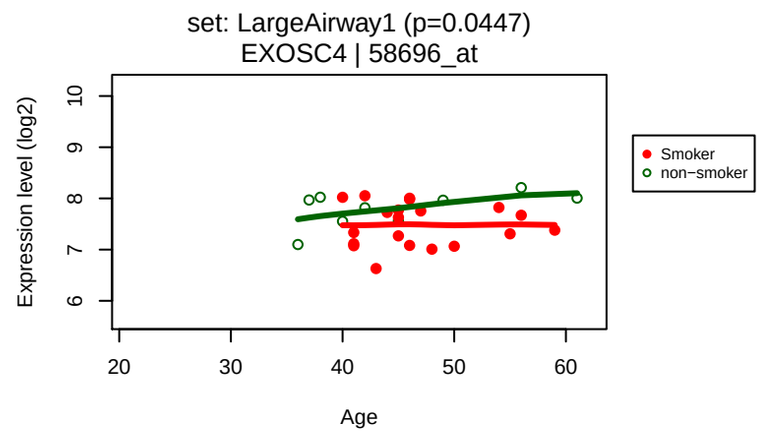

Gene expression of smokers and nonsmokers
gene: EXOSC4 | 58696_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0803
p=0.0213
p=0.0061
p=0.0447
p=0.0718
p=0.0256
p=0.1859
p=0.8814
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

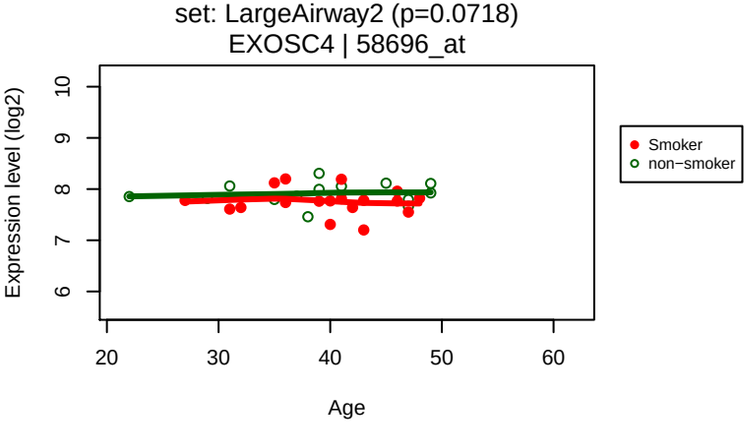

Gene expression ratio (smokers/nonsmokers)
gene: EXOSC4 | 58696_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0803
p=0.0213
p=0.0061
p=0.0447
p=0.0718
p=0.0256
p=0.1859
p=0.8814
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

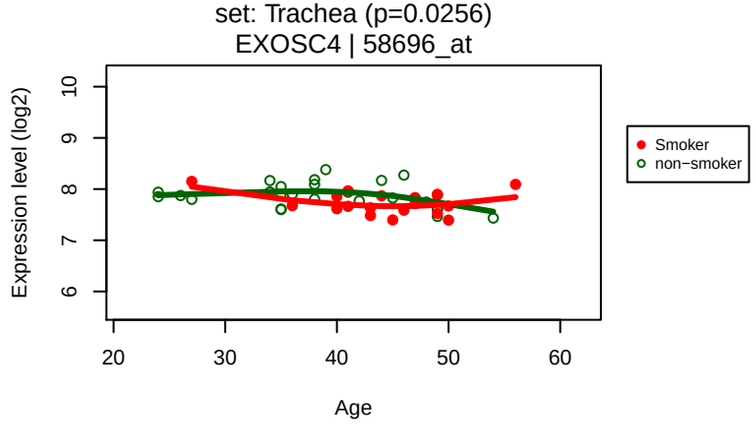

Gene expression ratio in Males
gene: EXOSC4 | 58696_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0238
p=0.3199
p=0.0634
p=0.6039
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: EXOSC4 | 58696_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5082
p=0.0876
p=0.0265
p=0.7689
SmallAirway2
LargeAirway2
Trachea
Macrophage2

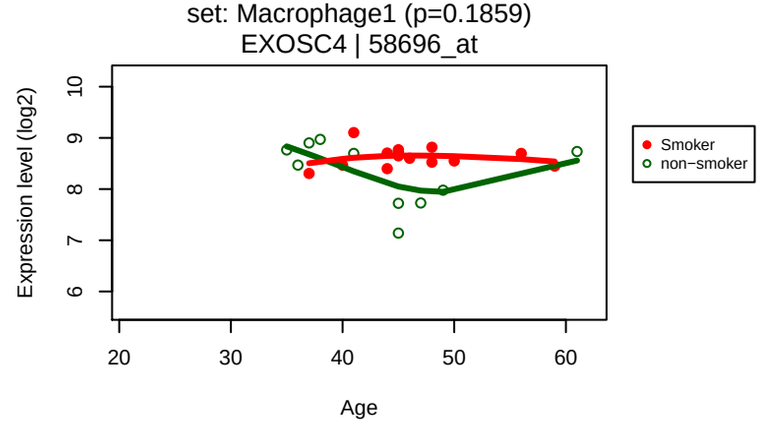

set: SmallAirway1 (p=0.0803)
EXOSC4 | 58696_at
Expression level (log2)
Age
Smoker
non−smoker

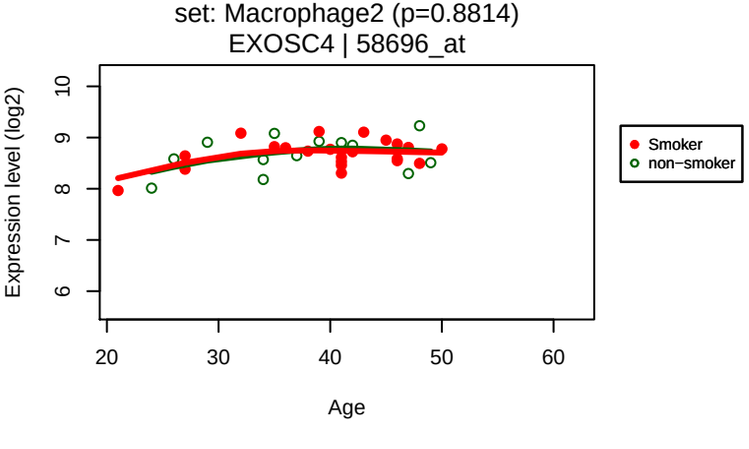

set: SmallAirway2 (p=0.0213)
EXOSC4 | 58696_at
Expression level (log2)
Age
Smoker
non−smoker

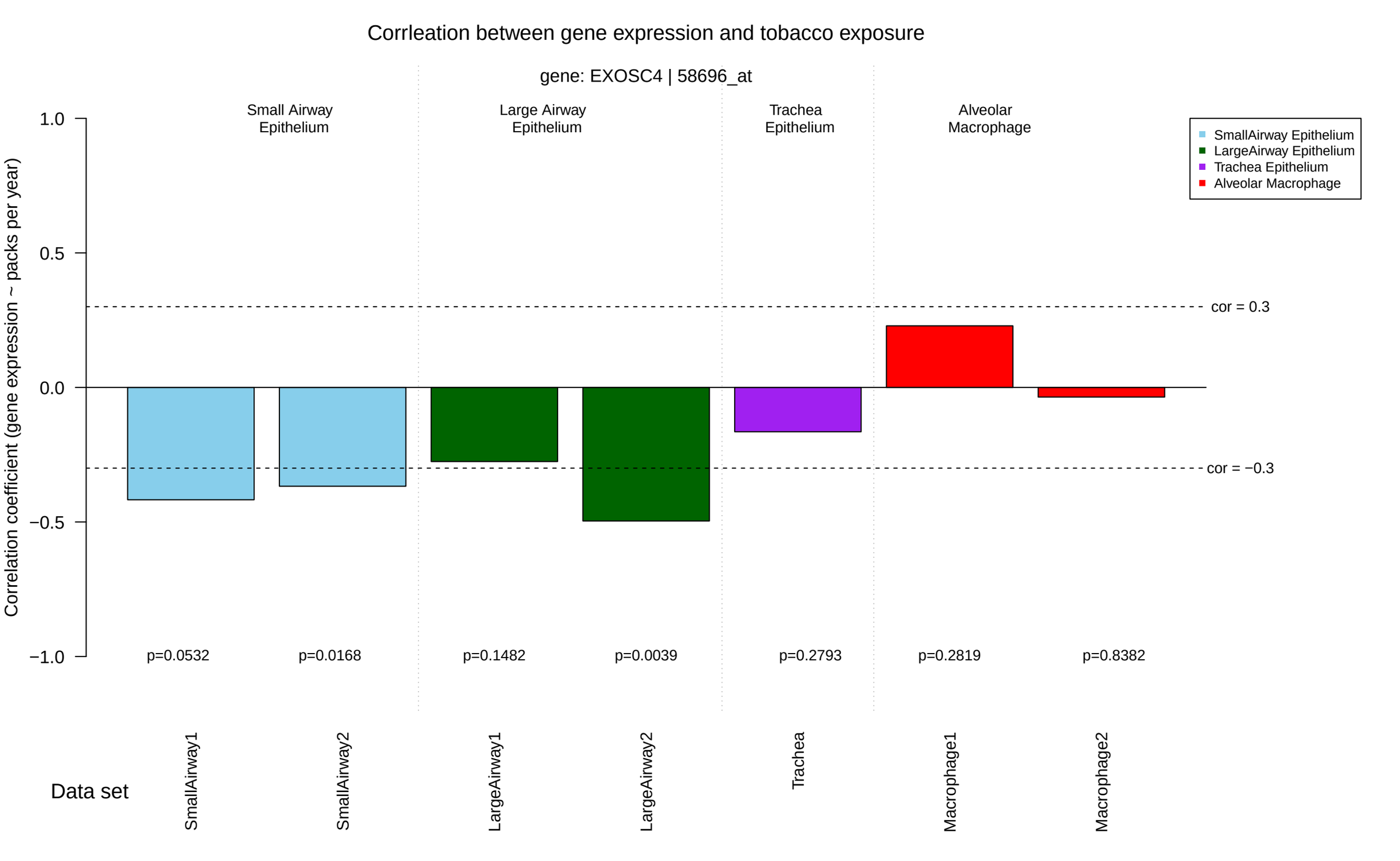

set: LargeAirway1 (p=0.0447)
EXOSC4 | 58696_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0718)
EXOSC4 | 58696_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0256)
EXOSC4 | 58696_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1859)
EXOSC4 | 58696_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8814)
EXOSC4 | 58696_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: EXOSC4 | 58696_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0532
p=0.0168
p=0.1482
p=0.0039
p=0.2793
p=0.2819
p=0.8382
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2